

A polynomial speed normalized health indicator for both incipient fault detection and prognosis of variable-speed wind turbine bearings

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ABSTRACT

The existing health indicators (HIs) are commonly available under stationary conditions. They may not accurately represent the degradation trajectory and cannot be applied to incipient fault detection (IFD) and remaining useful life (RUL) prediction under non-stationary conditions. Therefore, a new polynomial speed normalized HI (PSN-HI) is proposed to enable both IFD and RUL prediction of variable-speed wind turbine (WT) bearings. Firstly, the speed effect elimination method based on the correlation between the energy ratio of each decomposed component and speed is presented to partially eliminate the effects of speed on monitoring data. Then, a polynomial speed normalization method is proposed using the genetic algorithm and preset objective function to normalize the reconstructed data and further eliminate the influence of speed on vibration data. On this basis, the time-domain and frequency-domain HIs are generated through two mixture models. The WT bearing HI is constructed by linear weighting the two HIs. Finally, the comparative experiment results indicate that PSN-HI better characterizes the degradation process quantitatively and achieves both IFD and RUL prediction under varying speeds compared to the classical and state-of-the-art HIs.

1. Introduction

As a critical mechanical equipment converting wind into electrical energy, wind turbine (WT) has been successively equipped in many onshore and offshore wind farms (WFs). However, the incremental failure rates in WT components increase the operation and maintenance (O&M) costs while reducing the economic benefits of wind farms [1]. In addition, high-frequency O&M of key components increases the economic burden of WF. Therefore, incipient fault detection (IFD) and remaining useful life (RUL) prediction of WT components must be implemented. This implementation can reduce the enormous O&M costs for WF and improve the overall operational reliability of WT [2,3]. Health indicator (HI) construction, which can generate HI for quantitatively assessing the degenerate trajectory of the WT component, is indubitably a pivotal step to achieving the IFD and RUL prediction tasks. An appropriate HI can accurately detect the incipient fault and improve the accuracy of RUL prediction [4,5].

A series of HI construction methods have been explored in the past few years in the HI construction field. As the commonly used statistical parameter-based HIs, root mean square (RMS) and kurtosis can evaluate

the degradation process of mechanical component from the perspective of energy and impulsiveness, respectively [6]. Li et al. [7] proposed an adaptive first predicting time detection method based on kurtosis. The authors employed RMS and an improved exponential model to estimate the RULs of bearings. Several HIs based on sparsity measures and complexity measures, such as Gini index (GI) [8], negative entropy (NE) [9], approximate entropy (ApEn) [10], L2/L1 norm [11], and Hoyer measure (HM) [12], were utilized to assess the deterioration process of mechanical component. However, these HIs are susceptible to impulsive noise and non-stationary interference. Hence, they cannot effectively characterize the degradation trend of machinery. Although these statistical parameter-based HIs have specific physical meanings, they sometimes cannot comprehensively describe the degradation trend of mechanical component. This problem can mainly be attributed to HIs being constructed from a single aspect of statistical characteristic.

Data dimensionality reduction algorithms, such as principal component analysis (PCA), kernel principal component analysis (KPCA), and isometric mapping (ISOMAP), have been broadly employed to address the limitation of conventional statistical HIs by generating HIs for degradation assessment of machinery. Several statistical features are

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first extracted from the acquired monitoring data in these HI construction methods. Then, they are fused to construct a single HI to reflect the complicated machinery deterioration process. Mosallam et al. [13] adopted PCA to fuse time-domain and frequency-domain features and construct HI for machinery health monitoring. Wang et al. [14] proposed a Pearson-KPCA-based multi-domain feature fusion method to obtain fused features and characterize the bearing degradation trend. Qin et al. [15] constructed a gear HI from some time-domain and frequency-domain features using the ISOMAP method. Although these HIs can adequately describe the degradation process by fusing some statistical features, their performance is affected by some inapplicable features. In addition, prior knowledge is required to select additional effective features from the extracted statistical features.

Benefiting from the powerful feature mining capability of deep learning (DL) technology, some investigations on DL-based HI construction have been conducted in recent years. According to the manner of HI construction, the DL-based methods can be divided into supervised and unsupervised HI construction methods.

Regarding supervised HI construction approaches, Guo et al. [16] developed an HI construction method based on convolutional neural network (CNN) to generate bearing HI from raw monitoring data automatically. She et al. [17] designed a multi-channel deep CNN model to construct bearing HIs from multi-channel signals. Chen et al. [18] proposed an HI construction method based on recurrent neural networks, attention mechanisms, and linear HI labels.

For unsupervised HI construction methods, Xu et al. [19] constructed an intelligent HI based on a multiscale encoder-decoder model that can extract multi-scale features from input data. Ping et al. [20] established a log-normal-based variational auto-encoder (AE) network to extract degradation features for HI construction. Li et al. [21] presented an HI construction method based on canonical correlation analysis and AE to assess the health state of machinery quantitatively.

However, Most DL-based HIs are constructed under constant or nearly constant working conditions. In practical industrial applications, some machines, such as wind driven generators, industrial robots, and high-speed railways, often work under variable working conditions. These DL-based HI construction methods may be ineffective under variable operation conditions due to the coupling effect of operation speed and load on monitoring data. The working speed of equipment is usually measured in most cases, while the load is difficult to measure. Since the load is typically related to speed, this work mainly considers the HI construction under variable speed. The influence of load is also suppressed while the influence of speed is eliminated.

Some related investigations have been conducted in the last few years to alleviate or eliminate the impact of speed on monitoring signals. Zhou et al. [22] proposed a feature disentanglement model to disentangle the operating conditions and health states for bearing anomaly detection. Rao et al. [23] developed a speed-normalized auto-encoder to normalize the raw vibration signal through the speed normalization branch, removing the effect of speed variations. Yang et al. [24] introduced a normalization method to remove the amplitude modulation of speed conditions on vibration signals of a gearbox. Li et al. [25] built a generalized cumulative degradation model that considers the effect of speed variation for RUL prediction of WTs. The authors reduced the impact of the time-varying speed via a baseline transformation algorithm. Wei et al. [26] designed a rotating speed normalization method that considers the effect of the centrifugal force to normalize the raw monitoring signals. This method assumes that the monitoring signal amplitude positively correlates with the square of the rotating speed. The normalized data was obtained by dividing the acquired monitoring signal by the square of the rotating speed.

The analysis mentioned above reveals that the speed normalization method effectively eliminates the impact of speed variation on signal amplitude. The existing speed normalization method considers that the amplitude of the mechanical monitoring signal is related to the square of the corresponding speed. However, the amplitude of monitoring signal

in industrial scenario is not necessarily related to the square of speed. It may be related to the higher order of speed. The speed variation will also result in an energy ratio variation in the monitoring signal. In addition, the existing HIs can hardly achieve IFD and RUL prediction under variable speed.

Therefore, a new HI construction method is developed in this work for the degradation assessment of WT bearing under varying speed conditions. The speed effect elimination method based on energy ratio is proposed to partially remove the effect of speed on monitoring signals. The variational mode decomposition (VMD) is first applied to decompose the collected life-cycle signals into multiple components. Then, the correlation analysis between the energy ratio of each decomposed component and speed is conducted to reconstruct the life-cycle signal. Next, a polynomial speed normalization method is developed to eliminate the influence of speed on the amplitude of monitoring data. The coefficient of the speed polynomial is adaptively determined by the genetic algorithm (GA). Then, the HI of WT bearing is constructed by measuring the distribution discrepancies between the baseline distribution and distributions of monitoring data in time and frequency domains via Gaussian mixture model (GMM), exponential mixture model (EMM), and distribution contact ratio metric (DCRM). The constructed HI can characterize the degradation trend of WT bearing under variable speed conditions. IFD and RUL prediction of WT bearing can be achieved via the polynomial speed normalized HI (PSN-HI). Finally, the WT generator-bearing datasets validate the developed HI's effectiveness and superiority. The main contributions of this work are given as follows:

(1) A speed effect elimination approach based on energy ratio is presented to partially remove the influence of speed. The VMD is first employed to decompose the life-cycle signals. Then, the correlation between speed and each decomposed component's energy ratio is analyzed. Finally, the components weakly correlated with speed are used to reconstruct the life-cycle signal.

(2) The polynomial speed normalization method that considers the influence of speed on the amplitude of the monitoring signal is proposed for normalizing monitoring data. GA adaptively determines the coefficient of the speed polynomial, and the normalized monitoring signals are calculated by dividing the signal amplitude by the obtained speed polynomial.

(3) The WT bearing HI is constructed based on the normalized monitoring data from the perspective of distribution discrepancies. The effectiveness and superiority of the developed HI are experimentally validated. Compared with other HIs, the proposed HI suits both IFD and RUL prediction with a strong fault detection and prognosis ability.

2. Speed-normalized HI construction methodology

In the developed HI construction method, the speed effect elimination method based on energy ratio is first developed to partially eliminate the influence of speed on monitoring data. Afterward, polynomial speed normalization is proposed to remove the effects of speed on monitoring signals. Finally, the time-domain and frequency-domain HIs are constructed via GMM, EMM, and DCRM. The final HI of the WT bearing is constructed for health monitoring by linear weighting time-domain and frequency-domain HIs. The entire process of the developed HI construction method is visually shown in Fig. 1.

2.1. Speed effect elimination method based on energy ratio

As a completely non-recursive mode decomposition method, VMD can overcome the problems of end effect and mode component aliasing in the empirical mode decomposition method, characterized by a certain mathematical theoretical basis. Therefore, the VMD is applied to decompose the life-cycle monitoring signal and obtain multiple intrinsic mode functions (IMFs).

The life-cycle data of WT bearing is assumed as $x =$

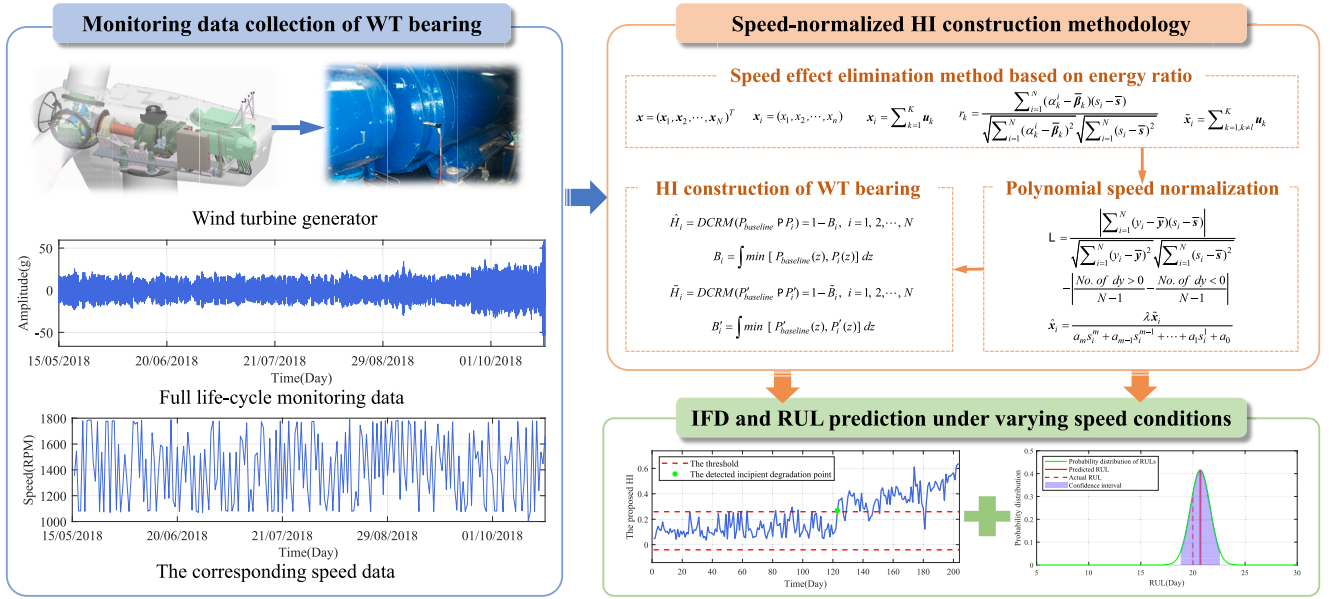


Fig. 1. The flowchart of the whole process of the proposed HI construction method.

$(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots, \mathbf{x}_N)^T$, $\mathbf{x}_i = (x_1, x_2, \dots, x_n)$, where N is the number of the life-cycle data and n is the data length of a single sampled data. The decomposition number of VMD is determined as K by the energy difference method [27]. The data $\mathbf{x}_i$ in the life-cycle signal is decomposed into a linear combination of modal components through VMD:

$$\mathbf{x}_i = \sum_{k=1}^K \mathbf{u}_k, \quad i = 1, 2, \dots, N \quad (1)$$

where $\mathbf{u}_k$ denotes the k th IMF component.

Multiple IMFs obtained by the VMD method can represent different components of monitoring signals. These IMFs are progressively increasing in frequency, from low to high. Hence, each IMF captures the vibration pattern or composition of the monitoring signal in the specific frequency range. The decomposition signals with strong speed correlation are selected through correlation analysis to analyze the relationship between different decomposition signals and speed. The energy of each IMF component is calculated as follows:

$$E_k = \sum_{i=1}^n |u_i|^2, \quad k = 1, 2, \dots, K \quad (2)$$

where E_k denotes the energy of the k th IMF component. Then, the relative importance of each IMF component α_k is quantified by calculating its energy proportion in all IMF components. The formula is as follows:

$$\alpha_k = \frac{E_k}{\sum_{k=1}^K E_k}, \quad k = 1, 2, \dots, K \quad (3)$$

Next, the energy ratio of each IMF component $\beta_k = (\alpha_k^1, \alpha_k^2, \dots, \alpha_k^N)$, $k = 1, 2, \dots, K$ is obtained via the life-cycle data $\mathbf{x}$. The Pearson correlation coefficient is applied to analyze the correlation between the energy ratio of each decomposed signal β_k and the speed $\mathbf{s} = (s_1, s_2, \dots, s_N)$ in the life cycle:

$$r_k = \frac{\sum_{i=1}^N (\alpha_k^i - \bar{\alpha}_k)(s_i - \bar{s})}{\sqrt{\sum_{i=1}^N (\alpha_k^i - \bar{\alpha}_k)^2} \sqrt{\sum_{i=1}^N (s_i - \bar{s})^2}}, \quad k = 1, 2, \dots, K \quad (4)$$

The l th IMF component with the highest correlation value is taken as the decomposition signal with a strong correlation. The other

decomposed signals are superimposed to reconstruct the monitoring signal $\mathbf{x}'_i$:

$$\mathbf{x}'_i = \sum_{k=1, k \neq l}^K \mathbf{u}_k, \quad i = 1, 2, \dots, N \quad (5)$$

Finally, the life-cycle data that partially eliminates the effect of speed is $\mathbf{x}' = (\mathbf{x}'_1, \mathbf{x}'_2, \dots, \mathbf{x}'_N)^T$.

2.2. Polynomial speed normalization

The speed effect elimination method based on energy variation can somewhat remove the partial effect of speed on monitoring data. However, the influence of speed on monitoring data cannot be comprehensively eliminated. A new polynomial speed normalization method is proposed in this work to further remove the influence of speed on monitoring signals. This method utilizes the speed signal to divide the original monitoring signal and eliminate the influence of the speed variation on the amplitude of monitoring data. The coefficient corresponding to each order in the speed polynomial is obtained using the GA method and the preset objective function defined as follows:

$$\mathcal{L} = \frac{|\sum_{i=1}^N (R_i - \bar{R})(s_i - \bar{s})|}{\sqrt{\sum_{i=1}^N (R_i - \bar{R})^2} \sqrt{\sum_{i=1}^N (s_i - \bar{s})^2}} - \left| \frac{\text{No. of } dR > 0}{N-1} - \frac{\text{No. of } dR < 0}{N-1} \right| \quad (6)$$

where $\mathbf{R}$ represents the RMS vector of monitoring signals after polynomial speed normalization. The specific order coefficients of the speed polynomial are obtained by minimizing the objective function through GA. The monitoring signal is divided by the speed polynomial to obtain the monitoring signals after speed normalization:

$$\hat{\mathbf{x}}_i = \frac{\lambda \mathbf{x}'_i}{a_m s_i^m + a_{m-1} s_i^{m-1} + \dots + a_1 s_i^1 + a_0}, \quad i = 1, 2, \dots, N \quad (7)$$

where $a_0, \dots, a_m$ are the coefficients of the speed polynomial; m is the order of the speed polynomial; λ is an adjustment factor controlling the scale of the monitoring signal. Finally, the monitoring data after polynomial speed normalization is $\hat{\mathbf{x}} = (\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \dots, \hat{\mathbf{x}}_N)^T$.

2.3. HI construction of WT bearing

GMM is first established based on the monitoring data after polynomial speed normalization to generate HIs in the time domain. The number of Gaussian components in GMM is set as C using Bayesian information criterion (BIC). The parameters of GMM are initialized by the K -means algorithm and iteratively updated through the expectation-maximum (EM) algorithm. The first M health data are utilized as the input data of GMM to estimate the baseline distribution $P_{baseline}$. The data distribution $\mathbf{P} = (P_1, P_2, \dots, P_N)$ for the monitoring data $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_N)^T$ are separately estimated via the GMM method. Afterward, the DCRM value $\hat{H}_i$ is obtained by $P_{baseline}$ and P_i as follows:

$$\hat{H}_i = DCRM(P_{baseline} \| P_i) = 1 - B_i, \quad i = 1, 2, \dots, N \quad (8)$$

where B_i denotes the area of the overlapping region between $P_{baseline}$ and P_i , and is computed via the integral formula as follows:

$$B_i = \int \min [P_{baseline}(z), P_i(z)] dz \quad (9)$$

The time-domain HI vector $\hat{\mathbf{H}} = (\hat{H}_1, \hat{H}_2, \dots, \hat{H}_N)$ is generated by the calculated DCRM values.

The time-domain HI can characterize the degradation process of WT bearing to a certain extent; however, it cannot fully represent the degradation trend of WT bearing. Therefore, the frequency-domain data is used to generate the frequency-domain HI of bearing. The FFT method is applied to obtain the FFT data of monitoring data $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_N)^T$. Then, the EMM is constructed while setting the number of exponential components as L through BIC. The EMM parameters are also updated by the EM algorithm. The first M health FFT data are used as the input data of EMM to estimate the baseline distribution $P'_{baseline}$, while the data distribution $\mathbf{P}' = (P'_1, P'_2, \dots, P'_N)$ for the entire FFT data $\hat{\mathbf{x}}$ are estimated via the constructed EMM. The DCRM value H_i is acquired by the following formula:

$$H_i = DCRM(P'_{baseline} \| P'_i) = 1 - \hat{B}_i, \quad i = 1, 2, \dots, N \quad (10)$$

where $\hat{B}_i$ denotes the area of the overlapping region between $P'_{baseline}$ and P'_i and is calculated as follows:

$$\hat{B}_i = \int \min [P'_{baseline}(z), P'_i(z)] dz \quad (11)$$

The frequency-domain HI vector $\mathbf{H} = (H_1, H_2, \dots, H_N)$ is obtained from the calculated DCRM values. Next, the bearing's PSN-HI is constructed by linear weighting the two HIs. The PSN-HI can be combined with time and frequency domain information to describe the degradation process of WT bearing.

2.4. IFD and RUL prediction of WT bearing

The 3σ interval is applied to determine the alarm threshold and accurately detect the incipient fault point (IFP) of the WT bearing. After finding the IFP, the first point of three consecutive points exceeding the alarm threshold is considered the true IFP. In contrast, the points without more than three successive points outside the alarm threshold are regarded as the pseudo-IFP.

As a variant of the long short-term memory (LSTM) network, the MLMA-Net model has been successfully applied to RUL prediction. The MLMA-Net can improve the memory capacity of LSTM by using memory augmented function and hierarchical mechanism. In this work, the MLMA-Net model is utilized to verify the effectiveness of PSN-HI in RUL prediction. In order to measure the uncertainty in RUL prediction, the dropout operation is employed to quantify the uncertainty of MLMA-Net in the testing stage. The cell structure of the MLMA-Net is displayed in Fig. 2. The additional information about the MLMA-Net model can be found in [28]. The forget gate f_t , input gate i_t , output gate o_t , and candidate state $\tilde{c}_t$ are calculated by the following formulas:

$$\begin{aligned} f_t &= \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f) \\ i_t &= \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i) \\ o_t &= \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o) \\ \tilde{c}_t &= \varphi(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c) \end{aligned} \quad (12)$$

where $\mathbf{x}_t$ denotes the input data at the current time; $\mathbf{h}_{t-1}$ represents the hidden state at previous time; $\sigma(\cdot)$ denotes the sigmoid function; $\varphi(\cdot)$ denotes the hyperbolic tangent function.

Based on the output values of forget gate and input gate, the attention distribution $\mathbf{A}_t$ is calculated as follows:

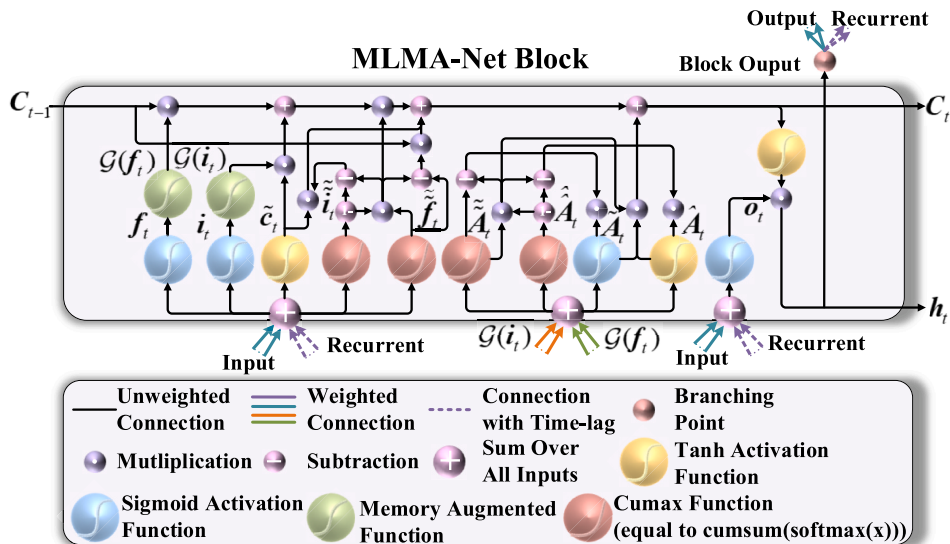


Fig. 2. The cell structure of MLMA-Net.

$$\begin{aligned}
\tilde{A}_t &= \sigma(W_A(f_t) + U_A(i_t) + b_A) \\
\hat{A}_t &= \varphi(W_A(f_t) + U_A(i_t) + b_A) \\
A_{d1} &= \text{softmax}(W_{\tilde{A}}(f_t) + U_{\tilde{A}}(i_t) + b_{\tilde{A}}) \\
A_{d2} &= \text{softmax}(W_{\hat{A}}(f_t) + U_{\hat{A}}(i_t) + b_{\hat{A}}) \\
\tilde{\tilde{A}}_t &= \text{cumsum}(A_{d1}) \\
\hat{\hat{A}}_t &= 1 - \text{cumsum}(A_{d2}) \\
\eta_{t1} &= \tilde{\tilde{A}}_t \odot \hat{\hat{A}}_t \\
\eta_{t2} &= \tilde{\tilde{A}}_t - \eta_{t1} \\
\eta_{t3} &= \hat{\hat{A}}_t - \eta_{t1} \\
A_t &= \eta_{t3} \odot \hat{\hat{A}}_t + \eta_{t1} \odot (\tilde{\tilde{A}}_t \odot \hat{\hat{A}}_t) + \eta_{t2} \odot \tilde{\tilde{A}}_t
\end{aligned} \tag{13}$$

where $\mathcal{Z}(\cdot)$ represents the memory-augmented function; $\tilde{A}_t$ denotes the attention ratio of gates; $\hat{A}_t$ is used to determine the output values of gates are positive or negative; $\tilde{\tilde{A}}_t$ and $\hat{\hat{A}}_t$ are the master attention gates; η_{t1} represents middle attention hierarchy; η_{t2} denotes high attention hierarchy; η_{t3} represents low attention hierarchy.

After that, the hidden state h_t of MLMA-Net is calculated as follows:

$$\begin{aligned}
\tilde{f}_t &= \text{softmax}(W_f x_t + U_f h_{t-1} + b_f) \\
\tilde{i}_t &= \text{softmax}(W_i x_t + U_i h_{t-1} + b_i) \\
\tilde{\tilde{f}}_t &= \text{cumsum}(\tilde{f}_t) \\
\tilde{\tilde{i}}_t &= 1 - \text{cumsum}(\tilde{i}_t) \\
\zeta_{t1} &= \tilde{\tilde{f}}_t \odot \tilde{\tilde{i}}_t \\
\zeta_{t2} &= \tilde{\tilde{f}}_t - \zeta_{t1} \\
\zeta_{t3} &= \tilde{\tilde{i}}_t - \zeta_{t1} \\
c'_t &= \mathcal{Z}(f_t) \odot c_{t-1} + \mathcal{Z}(i_t) \odot \tilde{c}_t \\
c_t &= \zeta_{t2} \odot c_{t-1} + \zeta_{t1} \odot c_t + \zeta_{t3} \odot \tilde{c}_t + A_t \\
h_t &= o_t \odot \varphi(c_t)
\end{aligned} \tag{14}$$

where $\tilde{f}_t$ is the master forget gate; $\tilde{i}_t$ is the master input gate; ζ_{t1} denotes the middle cell hierarchy; ζ_{t2} denotes the high cell hierarchy; ζ_{t3} represents the low cell hierarchy; c_t is the cell state at the current time.

The output of the MLMA-Net model is calculated using the hidden state h_t . The calculation formula is as follows:

$$\begin{aligned}
y_t^1 &= \text{Dropout}(h_t) \\
y_t^2 &= \text{Dropout}(\text{Relu}(w_1 y_t^1 + b_1)) \\
y_t^3 &= w_2 y_t^2 + b_2 \\
y_t &= w_3 (\text{Dropout}(y_t^3 + h_t)) + b_3
\end{aligned} \tag{15}$$

where y_t is the output of the MLMA-Net model; w and b are the weight and bias parameters of this model. The uniform initialization method is employed when training MLMA-Net to initialize weight parameters. The stochastic gradient descent method is used to update these parameters.

3. Experimental validation and analysis

In this section, the WT generator-bearing datasets are first introduced to validate the PSN-HI performance. Then, the bearing HIs are constructed using the developed HI construction method. The results of the PSN-HI are compared with other HIs. Finally, the PSN-HI is applied to the IFD and RUL prediction of WT bearing. The superiority of the PSN-HI for IFD detection and RUL prediction is verified through a comparison experiment with other HI construction methods.

3.1. WT generator bearing datasets description

The WT generator-bearing datasets are taken from an actual large wind field in the northern plain of China. The field sensor arrangement for this dataset is shown in Fig. 3. The accelerometers are installed at the drive and non-drive end of the generator. The monitoring data is collected and stored by a multi-channel data acquisition unit. The

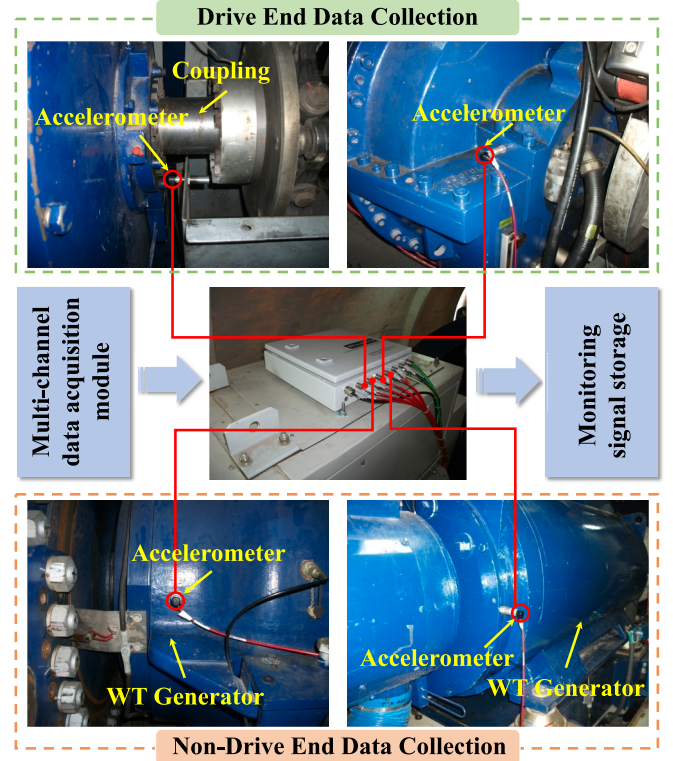


Fig. 3. The field sensor arrangement for WT generator bearing datasets.

sampling frequency was 25.6 kHz, and the sampling duration was 1.0 s during the life-cycle data collection process. In this work, the monitoring data of the drive end of the generator for #2 and #10 WTs is used to verify the performance of the proposed HI construction method under variable operating conditions. The time-domain waveforms of the full life-cycle data for WT generator bearings are shown in Fig. 4. The corresponding speed curves are shown in Fig. 5. The sample lengths of bearing life-cycle data are 204 and 225 for #2 and #10 WT, respectively.

During maintenance, serious wear and element slipping marks on bearing races were found, and the wear debris contaminated the lubricating grease. The defective bearing of the WT generator is shown in Fig. 6.

3.2. HI construction based on the proposed method

The drive end-bearing datasets of the two WTs are applied to verify the effectiveness of the proposed HI construction method. The decomposition numbers of VMD for #2 and #10 WTs are set to 3 and 5, respectively. After setting the decomposition number, the VMD decomposes the life-cycle signal. The correlation between the energy ratio of the decomposition signal and speed is analyzed. Then, the components weakly correlated with speed are employed to reconstruct the life-cycle signal and partially eliminate the effect of speed. Based on the reconstructed monitoring signals, the relative parameters of GA are used to optimize the coefficient of speed polynomial. The population numbers and the maximum number of iterations of GA for #2 and #10 WTs are set to 50, 20, 100, and 100. After determining the coefficient of the speed polynomial, the monitoring signals after speed normalization are obtained by dividing the amplitude of the monitoring signals by the speed polynomial. Thus, the influence of speed on the signal amplitude is further eliminated. The generator drive end-bearing HIs are constructed based on the normalized monitoring data using GMM and EMM. The first 50 normal samples under health state are used to estimate the baseline distribution. The constructed HIs are displayed in Fig. 7.

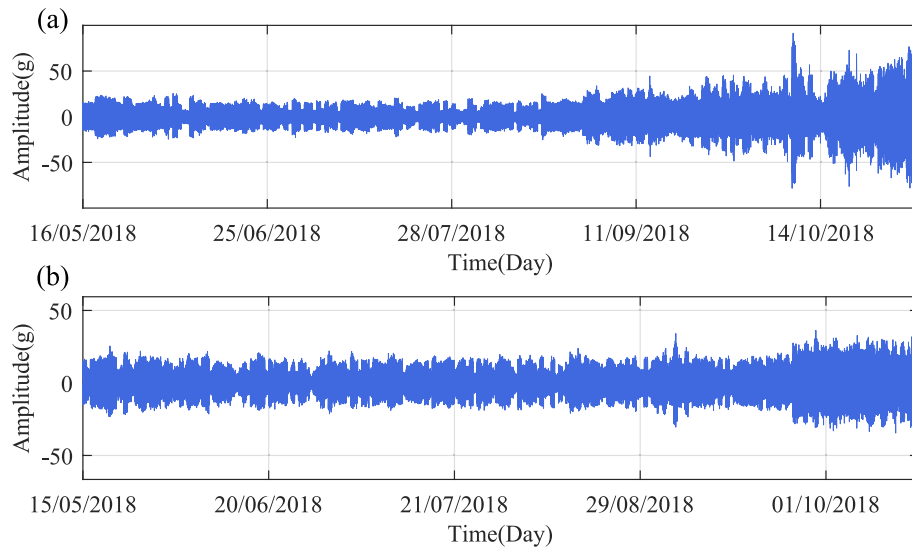


Fig. 4. The time-domain waveforms of the full life-cycle data for WT generator bearings. (a) #2 WT, (b) #10 WT.

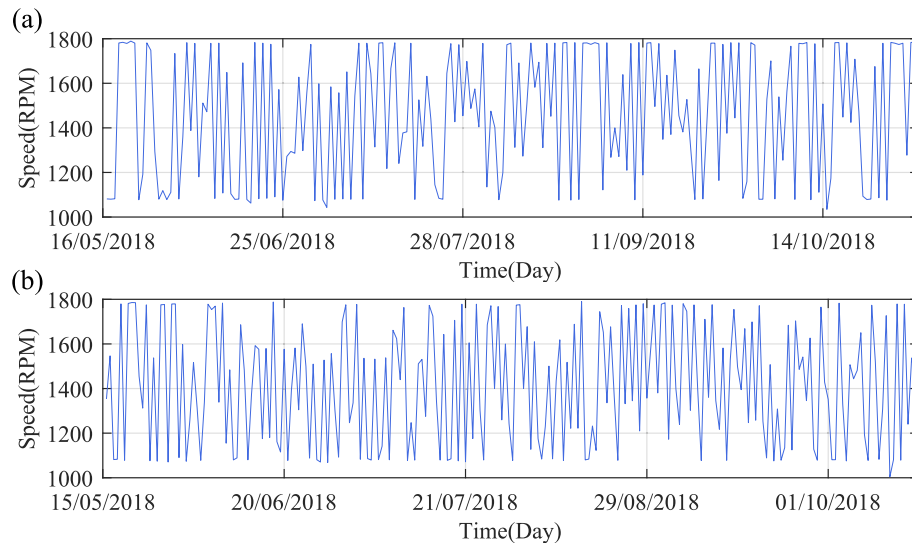


Fig. 5. The rotating speeds of the WT generators. (a) #2 WT, (b) #10 WT.



Fig. 6. The defected bearing of WT generator.

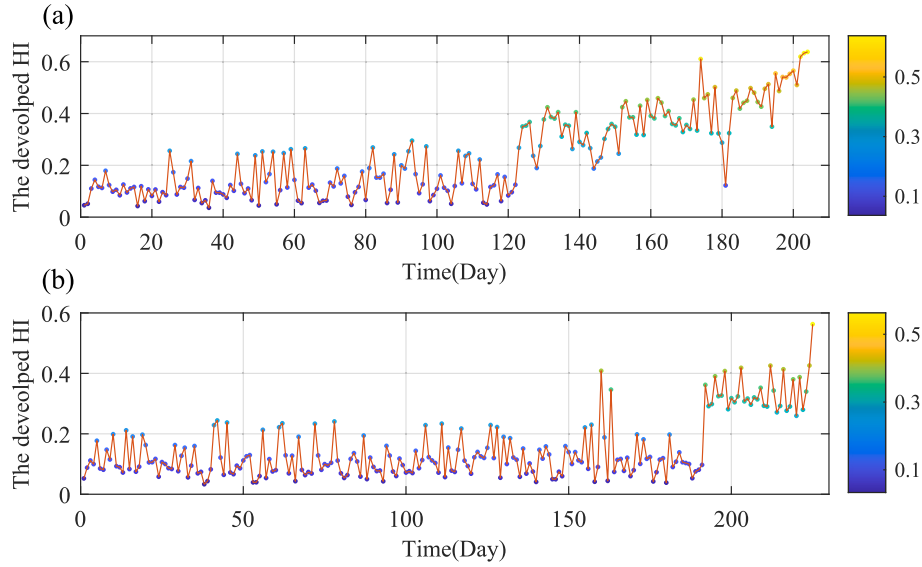


Fig. 7. The constructed HIs for WT bearings. (a) #2 WT, (b) #10 WT.

According to Fig. 7, the HI values are limited to the range [0, 1]. Hence, the dynamic degradation process of WT bearing under varying speed conditions can be quantitatively characterized. The HI value has smaller value in the normal stage, illustrating that the data distribution of the current monitoring data is similar to the baseline distribution. This observation indicates no defect in the WT bearing at this time. In the degradation stage, the HI value is much higher than that in the normal stage, demonstrating that the data distribution of the current monitoring data is quite different from the baseline distribution. This phenomenon shows that the WT bearing contains a serious defect. It can be concluded that the constructed HIs can effectively and quantitatively describe the degradation processes of WT bearings.

Several traditional and fusion HIs are applied to compare the proposed PSN-HI. In addition, GMM-based HI (GMM-HI), EMM-based HI (EMM-HI), GMM with a confidence value (GCV)-based HI (GCV-HI) [29] are also employed for further comparison. In order to verify the superiority of the polynomial speed normalization method, the HI without speed normalization (WSN-HI) and the HI with square speed normalization (SSN-HI) [26] are also used for contrastive analysis. In WSN-HI and SSN-HI, the two HIs are also constructed by linear weighting time-domain and frequency-domain HIs. Traditional HIs include RMS, kurtosis, GI [8], NE [9], ApEn [10], L2/L1 Norm [11], and HM [12]. Fusion HIs include PCA-based HI (PCA-HI), KPCA-based HI (KPCA-HI), and ISOMAP-based HI (ISOMAP-HI) [15]. These HIs are constructed from ten time-domain and frequency-domain features by PCA, KPCA, and ISOMAP. Four evaluation metrics, monotonicity (*Mon*), correlation (*Cor*), robustness (*Rob*), and composite index (*CI*), are utilized to evaluate the performance of HIs constructed by different methods. The weight coefficients of *Mon*, *Cor*, and *Rob* in *CI* are set to 0.4, 0.3, and 0.3, respectively [30]. Moreover, the promoting rate (PR) of the proposed PSN-HI compared to other HI is also utilized to demonstrate the advantage of the proposed PSN-HI. The PR is defined as follows:

$$PR = \left(\frac{CI_{Proposedmethod} - CI_{Comparativemethod}}{CI_{Comparativemethod}} \right) \times 100\% \quad (13)$$

The average evaluation results of different HIs (considering the WT generator-bearing datasets) are listed in Table 1. The proposed PSN-HI has the highest *CI* value of 0.531. In addition, the *CI* of the PSN-HI is 17.0 %, 45.5 %, 60.4 %, 46.7 %, 74.7 %, 51.7 %, 50.9 %, 21.0 %, 23.2 %, 16.5 %, 45.5 %, 8.4 %, 52.6 %, 7.1 % and 14.2 % higher than those of RMS, kurtosis, GI, NE, ApEn, L2/L1 Norm, HM, PCA-HI, KPCA-HI, ISOMAP-HI, GMM-HI, EMM-HI, GCV-HI, WSN-HI, and SSN-HI

Table 1

The average evaluation results of different HIs for #2 and #10 WTs.

Method	<i>Mon</i>	<i>Cor</i>	<i>Rob</i>	<i>CI</i>	PR
RMS	0.144	0.544	0.778	0.454	↑17.0 %
Kurtosis	0.124	0.451	0.599	0.365	↑45.5 %
GI	0.094	0.419	0.560	0.331	↑60.4 %
NE	0.124	0.454	0.586	0.362	↑46.7 %
ApEn	0.096	0.358	0.528	0.304	↑74.7 %
L2/L1 Norm	0.108	0.442	0.579	0.350	↑51.7 %
HM	0.114	0.442	0.579	0.352	↑50.9 %
PCA-HI	0.133	0.576	0.709	0.439	↑21.0 %
KPCA-HI	0.118	0.581	0.700	0.431	↑23.2 %
ISOMAP-HI	0.136	0.586	0.752	0.456	↑16.5 %
GMM-HI	0.169	0.349	0.644	0.365	↑45.5 %
EMM-HI	0.050	0.706	0.859	0.490	↑8.4 %
GCV-HI	0.171	0.277	0.656	0.348	↑52.6 %
WSN-HI	0.148	0.649	0.808	0.496	↑7.1 %
SSN-HI	0.173	0.603	0.715	0.465	↑14.2 %
PSN-HI	0.230	0.669	0.793	0.531	/

respectively. The *Mon* result of the PSN-HI is higher than that of all comparison HIs. Moreover, *Cor* and *Rob* results are also higher than other comparison HIs, except EMM-HI and WSN-HI. Therefore, the PSN-HI is characterized by the best comprehensive performance and can quantitatively describe the dynamic degradation process of WT generator bearing under varying speeds. This indicates that the speed effect elimination method based on energy ratio and polynomial speed normalization can effectively eliminate the effect of speed variation to construct HIs of WT generator bearings.

3.3. IFP detection of WT bearing with the PSN-HI

The first 50 HI values under the normal stage are used to calculate the mean value and standard deviation value in the 3σ interval $[\mu - 3\sigma, \mu + 3\sigma]$. The detection results with the proposed HIs for #2 and #10 WTs are shown in Fig. 8. The IFPs can be accurately detected, and the IFPs of #2 and #10 WT bearings are at positions 123 and 192, respectively. The fault characteristic frequencies (FCFs) of WT bearings are calculated when the fundamental frequency of the shaft is equal to 1 Hz to confirm the detection results, as shown in Table 2. The waveforms and spectrums of monitoring signal at IFP are shown in Figs. 9 and 10. The FCFs of WT bearing are prominently observed in the obtained spectrums, validating that bearing incipient defect has occurred at the detected position and verifying the accuracy of the IFD based on the

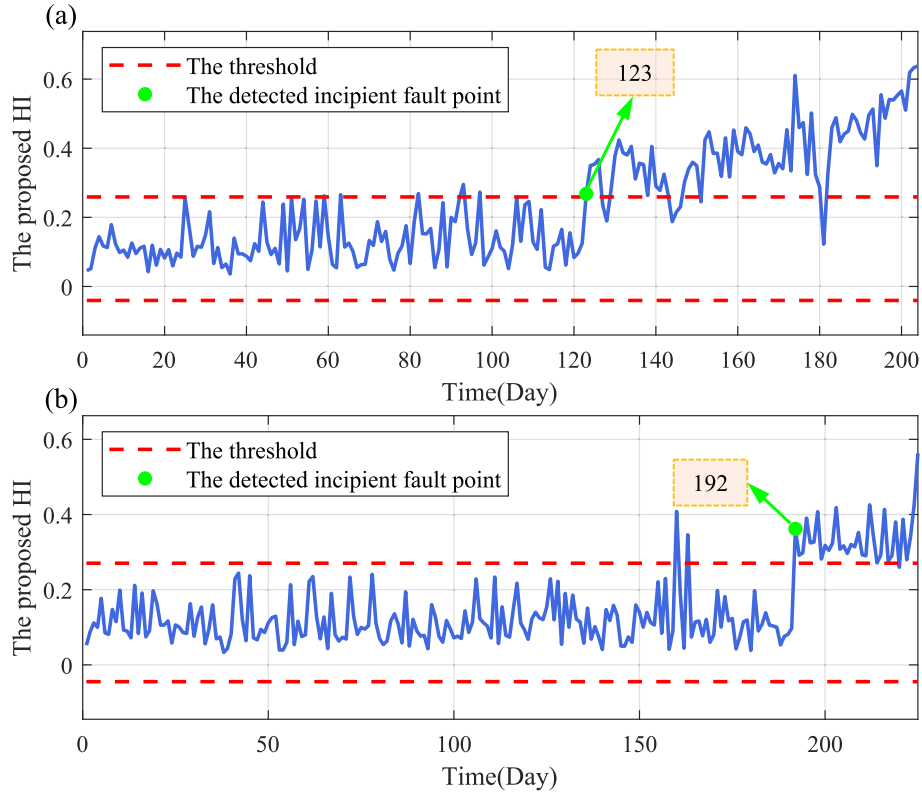


Fig. 8. The IFD results with the proposed HIs. (a) #2 WT, (b) #10 WT.

Table 2
The FCFs of WT bearing.

Bearing type	Rotating f_r (Hz)	Outer race f_o (Hz)	Inner race f_i (Hz)	Ball f_b (Hz)	Cage f_c (Hz)
6334 M/C3	1	3.16	4.84	4.56	0.4

proposed HI. This verification illustrates that the developed HI, through polynomial speed normalization, is feasible and effective in IFD of WT bearing under varying speed conditions.

Comparative HIs are applied to detect the IFPs of WT bearings. The detection results based on the comparative HIs for #2 WT are shown in Fig. 11. It can be intuitively seen that the constructed HIs based on GI,

NE, ApEn, L2/L1 Norm, HM, GMM, GCV, and SSN all have obvious numerical fluctuation phenomenon in the health state stage. This phenomenon may trigger false alarms in the process of health status monitoring. In addition, the detected IFPs of different HIs for #2 WT are listed in Table 3. The constructed HIs based on GI, NE, ApEn, L2/L1 Norm, and HM are all failed to detect the IFP of bearing. The IFPs detected by RMS, Kurtosis, PCA-HI, KPCA-HI, ISOMAP-HI, GMM-HI, and GCV-HI are delayed. The above HIs are insensitive to the initial fault due to the amplitude effect caused by speed variation. The IFP detected by SSN-HI is also delayed. This is mainly because the square normalization method only considers the square of speed, and does not take into account the higher order of speed. Although the IFPs detected by EMM-HI and WSN-HI are same as the one detected by PSN-HI, both EMM-HI

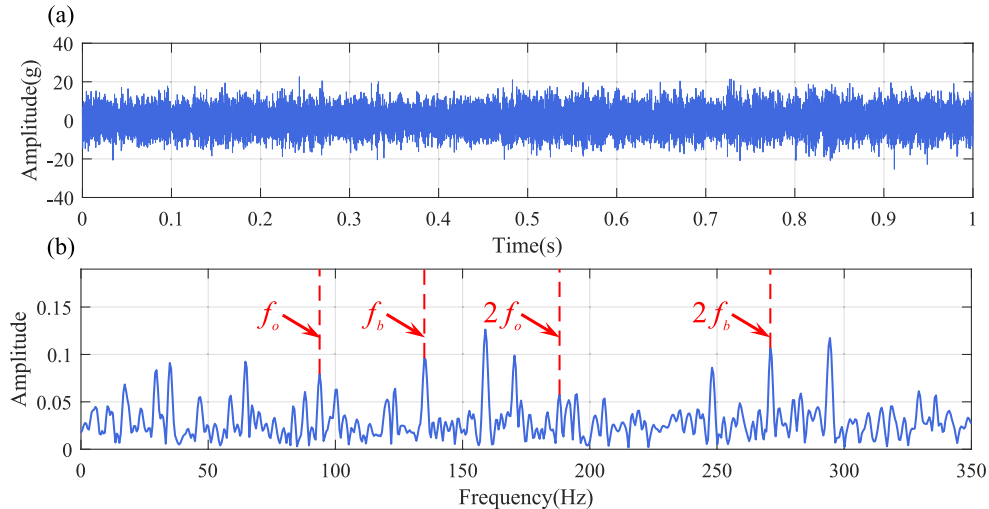


Fig. 9. The waveform and spectrum of monitoring data at position 123 for #2 WT (a) The waveform of monitoring data, (b) The spectrum of monitoring data.

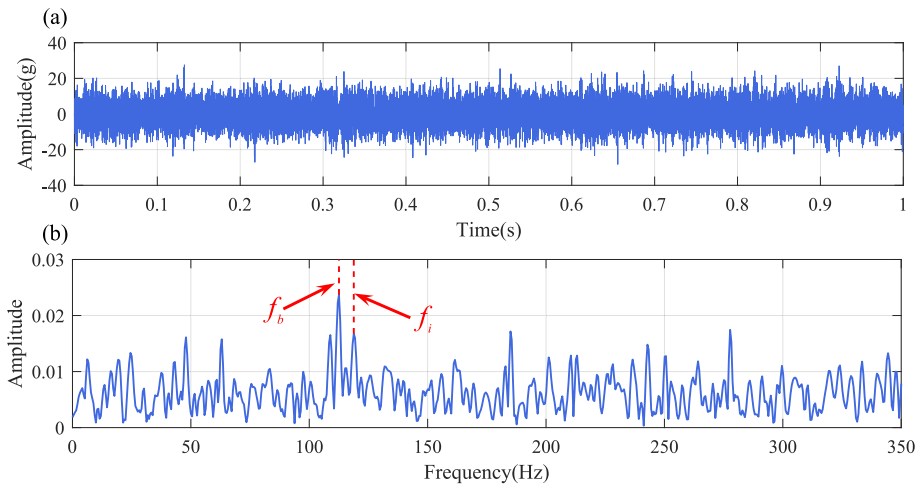


Fig. 10. The waveform and spectrum of monitoring data at position 192 for #10 WT (a) The waveform of monitoring data, (b) The spectrum of monitoring data.

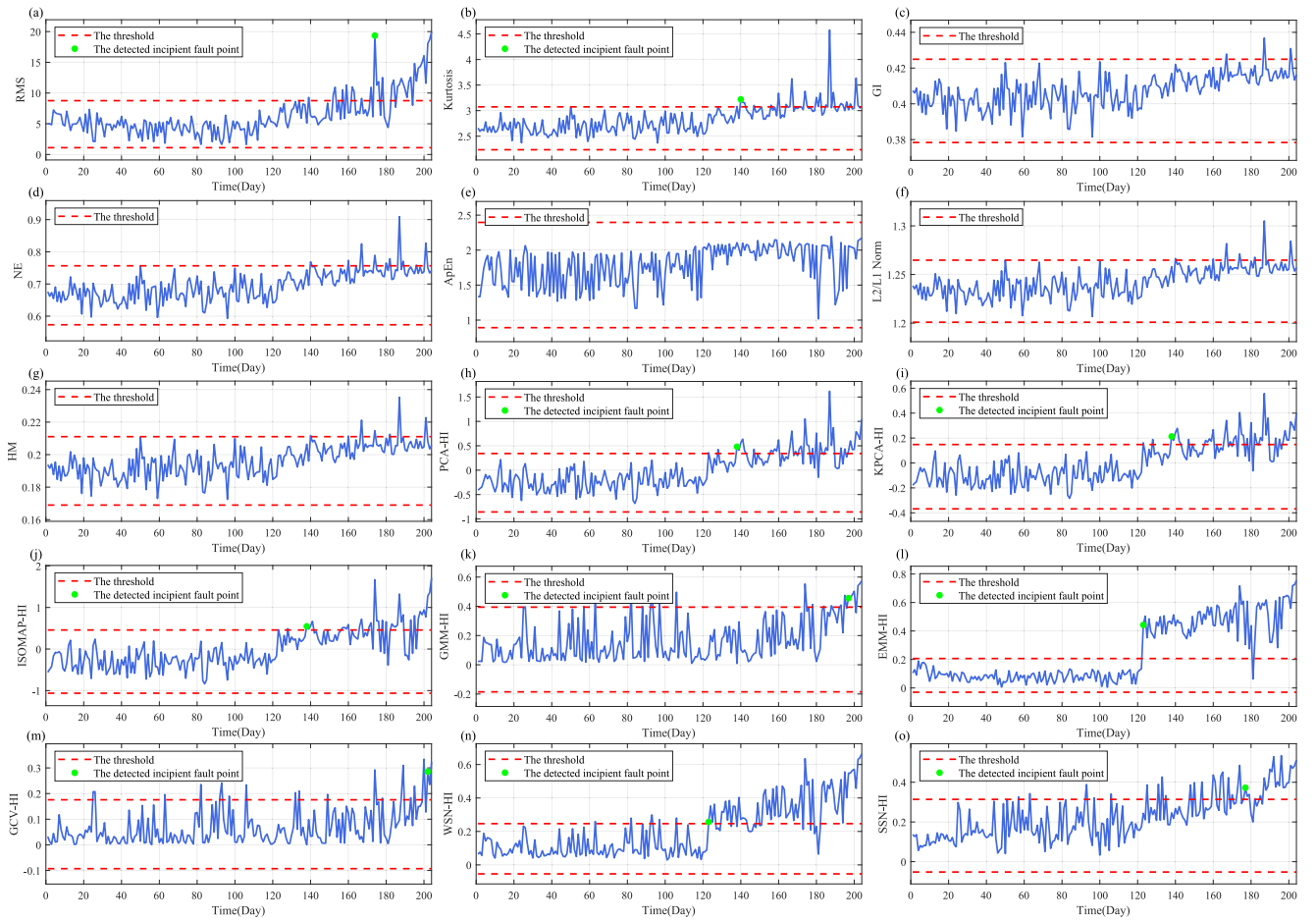


Fig. 11. The IFD results based on the comparative HIs for #2 WT (a) RMS, (b) Kurtosis, (c) GI, (d) NE, (e) ApEn, (f) L2/L1 Norm, (g) HM, (h) PCA-HI, (i) KPCA-HI, (j) ISOMAP-HI, (k) GMM-HI, (l) EMM-HI, (m) GCV-HI, (n) WSN-HI, (o) SSN-HI.

Table 3

The IFD results of different HIs for #2 WT.

Method	RMS	Kurtosis	GI	NE	ApEn	L2/L1 Norm	HM	PCA-HI
IFP	174	140	—	—	—	—	—	138
Method	KPCA-HI	ISOMAP-HI	GMM-HI	EMM-HI	GCV-HI	WSN-HI	SSN-HI	PSN-HI
IFP	138	138	197	123	202	123	177	123

and WSN-HI are characterized by a large fluctuation after IFP, which is not conducive to RUL prediction.

For #10 WT, the detection results based on the comparative HIs are shown in Fig. 12. It can be observed from Fig. 12 that most of the comparative HIs cannot effectively characterize the degradation process of WT bearing. The constructed HIs by RMS, Kurtosis, GI, NE, ApEn, L2/L1 Norm, HM, PCA-HI, KPCA-HI, GMM-HI, and GCV-HI failed to detect the IFP of bearing. The IDPs detected by ISOMAP-HI, EMM-HI, WSN-HI, and SSN-HI are respectively 223, 192, 192, and 200. They occur later or are the same as the IDP obtained by the proposed PSN-HI. It can be concluded that the proposed PSN-HI has a stronger IFD ability and achieves higher IFD accuracy than the conventional and state-of-the-art HIs.

3.4. RUL prediction of WT bearing with the PSN-HI

The constructed HIs for #2 and #10 WTs are applied to estimate the RULs of WT bearings through the established MLMA-Net model. The first 184 HI values train the MLMA-Net model for #2 WT. Then, the last 20 HI values are predicted by the trained MLMA-Net. The first 205 HI values are utilized to build the training set for #10 WT. Then, the MLMA-Net is trained to estimate the remaining 20 HI values.

One hundred parallel prediction experiments were conducted to obtain 100 prediction results and ensure the fairness and reliability of the RUL prediction results. The average value of these prediction results was taken as the final RUL of WT bearing. In the prediction phase, the Monte-Carlo dropout operation was enabled to quantify the uncertainty of the prediction model. The actual RULs, predicted RULs, 95 %

confidence interval, and probability distribution of RULs for #2 and #10 WTs are displayed in Fig. 13.

The predicted results demonstrate that the RULs obtained by the MLMA-Net model are very close to the actual RULs, and the predicted RULs are within the 95 % confidence interval. According to the probability distribution of RULs, the dispersion degree of the probability distribution of RUL is relatively concentrated, illustrating small uncertainty. Therefore, it can be concluded that the PSN-HI is appropriate for RUL prediction, as it can obtain accurate RUL prediction results based on the MLMA-Net model.

Other comparative HIs are employed for comparative analysis to verify the superiority of the proposed PSN-HI in RUL prediction. For #2 WT, RMS, Kurtosis, GI, NE, L2/L1 Norm, HM, PCA-HI, KPCA-HI, ISOMAP-HI, GMM-HI, EMM-HI, and GCV-HI, WSN-HI, and SSN-HI are utilized for comparative analysis. Since the HI constructed by ApEn is not characterized by an obvious degradation trend, it is not considered for comparative analysis in this section. For #10 WT, RMS, PCA-HI, KPCA-HI, ISOMAP-HI, GMM-HI, EMM-HI, WSN-HI, and SSN-HI are used for comparative analysis. The Kurtosis, GI, NE, ApEn, L2/L1 Norm, HM, and GCV-HI show no degradation trend. Therefore, these HIs are not employed for comparison. Mean absolute error (MAE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), root mean square error (RMSE), and Score are utilized to evaluate the predictive performance of different HIs quantitatively. The evaluation results of various HIs in RUL prediction are listed in Tables 4 and 5.

Tables 4 and 5 show that the proposed HI is characterized by the lowest prediction error and the highest Score value. For #2 WT, the PSN-

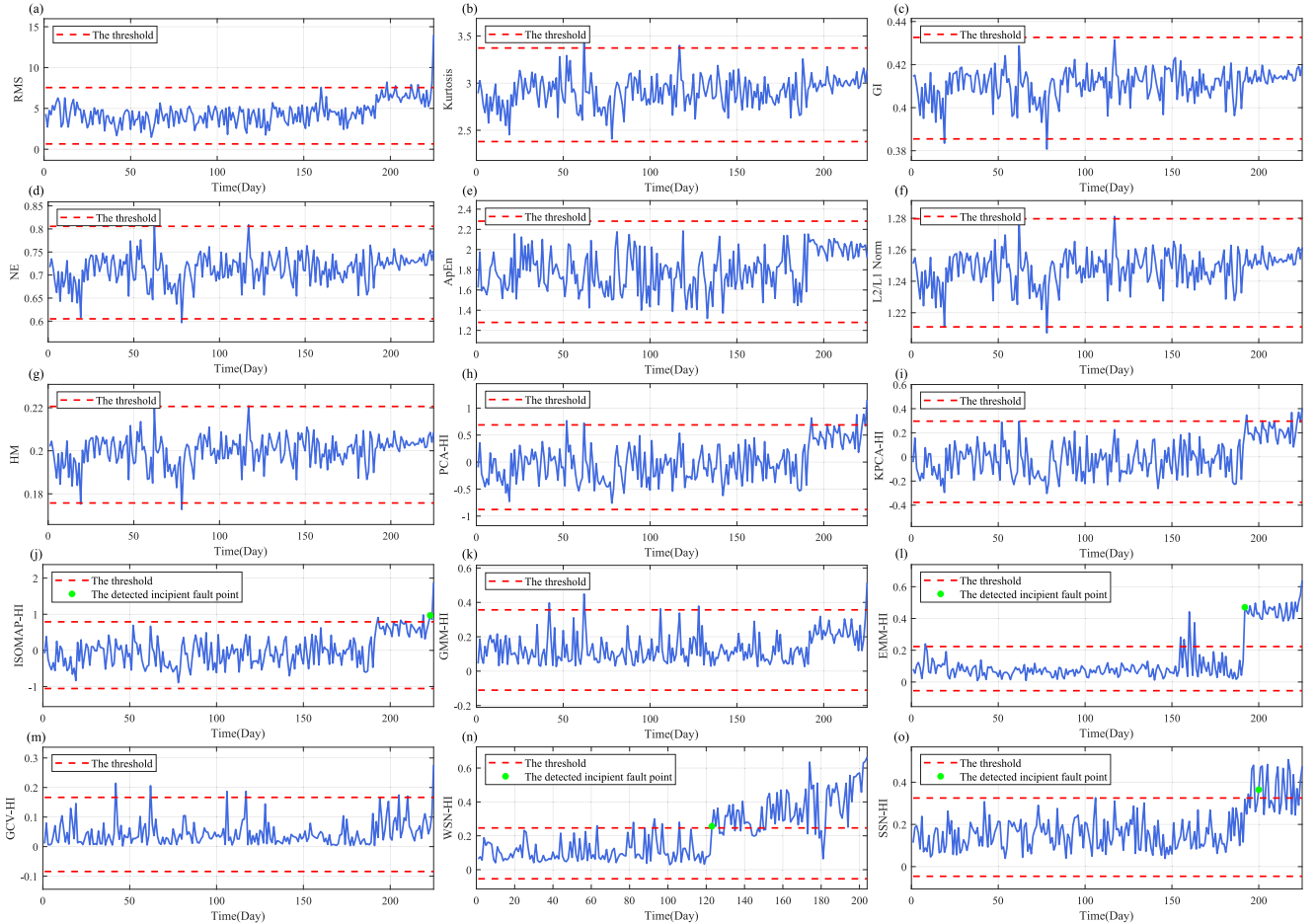


Fig. 12. The IFD results based on the comparative HIs for #10 WT (a) RMS, (b) Kurtosis, (c) GI, (d) NE, (e) ApEn, (f) L2/L1 Norm, (g) HM, (h) PCA-HI, (i) KPCA-HI, (j) ISOMAP-HI, (k) GMM-HI, (l) EMM-HI, (m) GCV-HI, (n) WSN-HI, (o) SSN-HI.

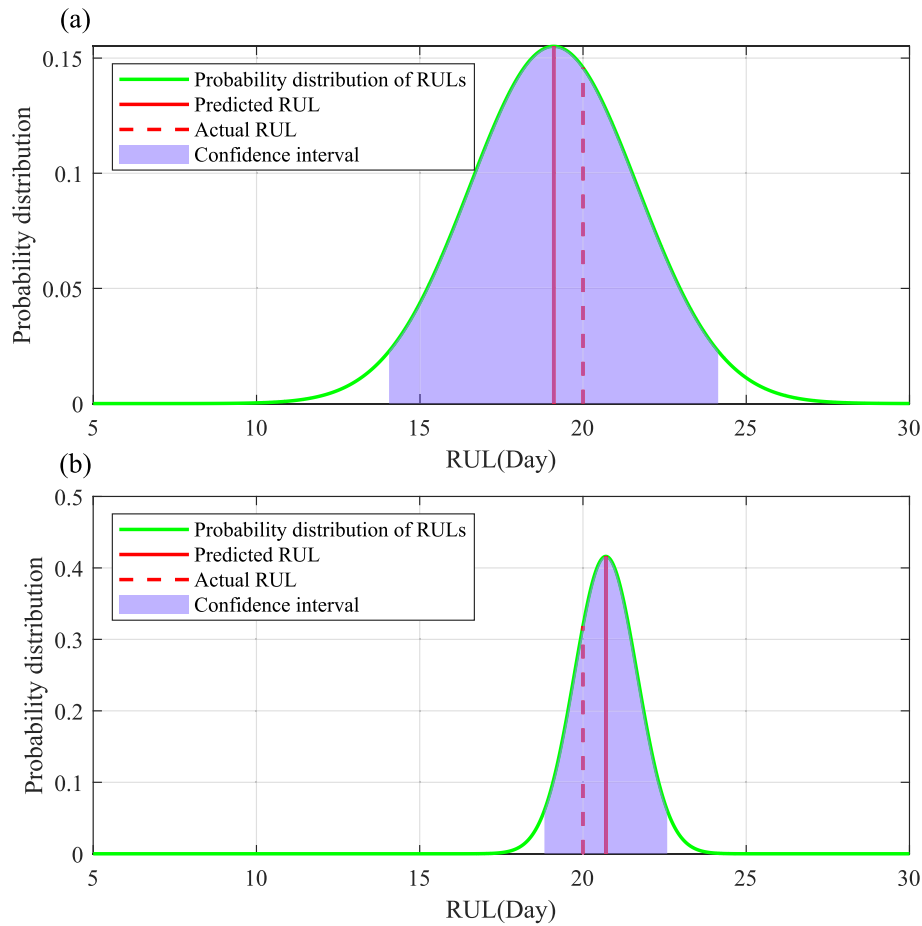


Fig. 13. The prediction results with uncertainty assessment. (a) #2 WT, (b) #10 WT.

Table 4

The evaluation results of different HIs for #2 WT.

Method	MAE	NRMSE	MAPE	RMSE	Score
RMS	6.830	0.461	34.15 %	7.138	0.246
Kurtosis	9.710	0.998	48.55 %	10.392	0.215
GI	10.890	0.703	54.45 %	13.388	0.176
NE	8.560	0.381	42.80 %	10.170	0.182
L2/L1 Norm	6.100	0.335	30.50 %	8.745	0.158
HM	7.730	0.557	38.65 %	9.225	0.286
PCA-HI	10.670	0.401	53.35 %	11.907	0.075
KPCA-HI	6.180	0.309	30.90 %	8.022	0.072
ISOMAP-HI	13.920	0.962	69.60 %	14.588	0.093
GMM-HI	10.930	0.578	54.65 %	17.324	0.454
EMM-HI	4.720	0.312	23.60 %	4.771	0.445
GCV-HI	10.750	0.393	53.75 %	12.060	0.018
WSN-HI	5.810	0.379	29.05 %	9.768	0.333
SSN-HI	6.650	0.465	33.25 %	7.499	0.350
PSN-HI	2.640	0.142	13.20 %	2.733	0.495

Table 5

The evaluation results of different HIs for #10 WT.

Method	MAE	NRMSE	MAPE	RMSE	Score
RMS	6.160	0.243	30.80 %	6.348	0.025
PCA-HI	4.470	0.290	22.35 %	4.500	0.463
KPCA-HI	4.980	0.332	24.90 %	4.982	0.422
ISOMAP-HI	4.640	0.308	23.20 %	4.726	0.454
GMM-HI	3.690	0.272	18.45 %	6.451	0.212
EMM-HI	2.010	0.091	10.05 %	2.012	0.249
WSN-HI	2.400	0.112	12.00 %	2.518	0.210
SSN-HI	8.870	0.842	44.35 %	11.072	0.270
PSN-HI	0.960	0.056	4.80 %	1.104	0.737

HI is better than the traditional HIs, such as RMS, Kurtosis, GI, NE, L2/L1 Norm, and HM. The predictive performance of the PSN-HI is also better than those of PCA-HI, KPCA-HI, ISOMAP-HI, GMM-HI, EMM-HI, GCV-HI, WSN-HI, and SSN-HI. For #10 WT, the proposed PSN-HI is better than all comparison methods. The above results demonstrate that the PSN-HI is more suitable for RUL prediction of WT bearing and can improve the accuracy of RUL prediction. It can be concluded that the speed effect elimination method based on energy ratio and polynomial speed normalization is able to effectively eliminate the influence of speed on monitoring data for constructing HI with excellent performance. In addition, the constructed PSN-HI has better predictive performance than other conventional and state-of-the-art HIs.

4. Conclusions

This paper presented a new PSN-HI to enable the IFD and RUL prediction of WT bearings under variable speed conditions. The speed effect elimination method based on energy ratio was first developed. This method employed the VMD to decompose the original life-cycle signal into multiple components. Then, the correlation between the energy ratio of each component and speed was analyzed to reconstruct monitoring data and partially remove the effects of speed on vibration signals. The polynomial speed normalization method was developed based on the reconstructed life-cycle signals by employing the GA and the objective function related to speed. The normalized vibration signals were acquired by dividing the reconstructed signal with the speed polynomial. The time-domain and frequency-domain HIs were generated using GMM and EMM based on the obtained results, and the final HI of WT bearing was constructed by linear weighting the two HIs. Finally, the effectiveness and superiority of the proposed PSN-HI were validated

by the WT generator-bearing datasets. Moreover, the PSN-HI was utilized for IFD and RUL prediction. The experimental analysis results illustrate that the PSN-HI can describe the nonlinear deterioration process and is superior to conventional and state-of-the-art HIs. In addition, the PSN-HI is suitable for IFD and RUL prediction under variable speed conditions. The PSN-HI can obtain more accurate fault detection and prognosis results than RMS, Kurtosis, GI, NE, L2/L1 Norm, HM, PCA-HI, KPCA-HI, ISOMAP-HI, GMM-HI, EMM-HI, GCV-HI, WSN-HI, and SSN-HI.

However, it is worth mentioning that the proposed PSN-HI is only applicable to the IFD and RUL prediction of WT bearings under time-varying speed operation conditions. To address this weakness, the future work will be dedicated to developing an unsupervised HI construction method under variable speed and load conditions to achieve the accurate IFD and RUL prediction of WT bearings.

CRedit authorship contribution statement

Dingliang Chen: Writing – original draft, Methodology. **Yi Wang:** Visualization, Data curation. **Yi Chai:** Validation, Formal analysis. **Yuejian Chen:** Software, Investigation. **Yi Qin:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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